

KNOCK/ OUT/MATH

SAMPLE NOTES: Problems in Real-World Contexts (PRWC)

Secondary G3 Math



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The new MOE syllabus (4052) states:

AIMS

The O-Level Mathematics syllabus aims to enable all students to:

- acquire mathematical concepts and skills for continuous learning in mathematics and to support learning in other subjects
- develop thinking, reasoning, communication, application and metacognitive skills through a mathematical approach to problem-solving
- connect ideas within mathematics and between mathematics and other subjects through applications of mathematics
- build confidence and foster interest in mathematics.

SCHEME OF ASSESSMENT

Paper	Duration	Description	Marks	Weighting
Paper 1	2 hours 15 minutes	There will be about 26 short answer questions. Candidates are required to answer all questions.	90	50%
Paper 2	2 hours 15 minutes	There will be 9 to 10 questions of varying marks and lengths. The last question in this paper will focus specifically on applying mathematics to a real-world scenario. Candidates are required to answer all questions.	90	50%

PROBLEMS IN REAL-WORLD CONTEXTS

Notwithstanding the presentation of the topics in 3 separate strands in the syllabus document, it is envisaged that some examination questions (including the extended problem involving real-world contexts at the end of Paper 2) may integrate ideas from more than one topic.

Problems in real-world contexts may be based on contexts:

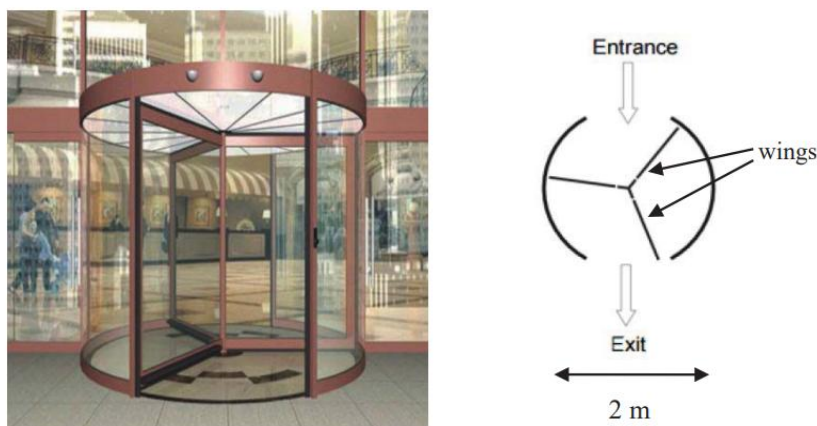
- In everyday life (including travel/excursion plans, transport schedules, sports and games, recipes, floor plans, navigation etc.)
- Involving personal and household finance (including simple and compound interest, taxation, instalments, utilities bills, money exchange, etc.)

These problems may also require:

- Interpreting and analysing data from tables and graphs, including distance-time and speed-time graphs;
- Interpreting the solution in the context of the problem.

Application of Inequalities in Real World context (Secondary 2 Topic)

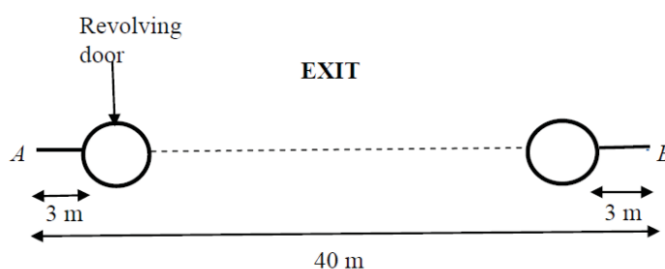
The photograph below shows a three-wing revolving door.



The three wings divide the partial circular-shaped enclosure into three equal parts. The internal diameter of this enclosure is 2 metres. The diagram below shows the top view of this revolving door.

- (a) What is the size in degrees of the angle formed by two wings?
- (b) The door makes 4 complete rotations in a minute. There are only allowed a maximum of two people in each of the three wings. Write down the rate at which the people leave the doors in the form k people/minute, where k is a whole number.

Given that the revolving doors are to be built at the exit of a shopping mall and along a straight path AB of length 40 m.

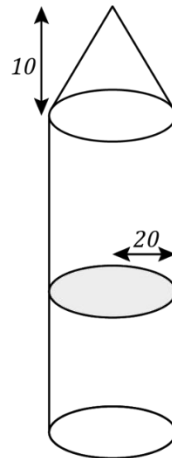


Some important information is stated below:

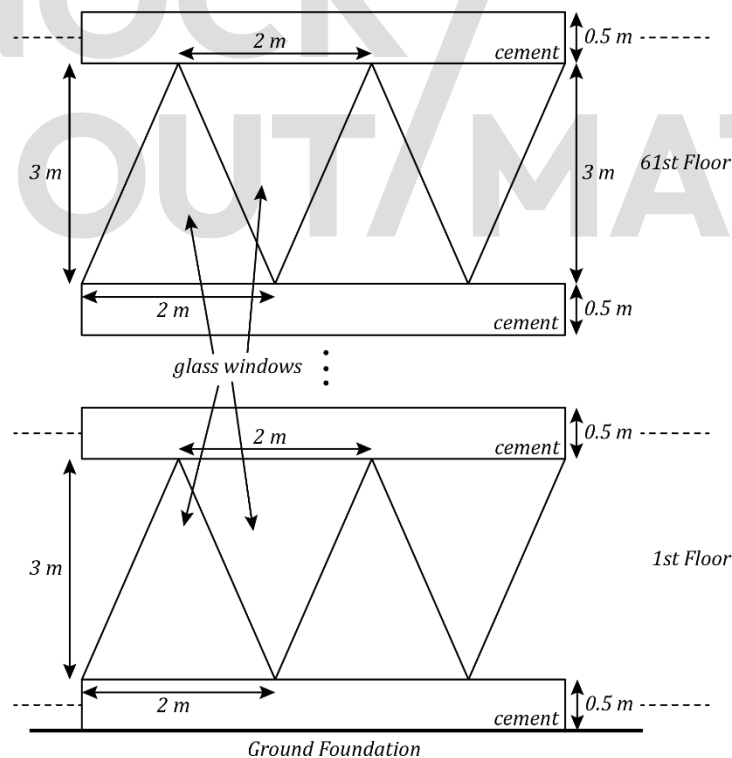
1. The maximum capacity of the shopping mall is 3000 people.
 2. According to the fire safety regulations, the maximum time limit for all the shoppers to be able to exit the mall is 30 minutes.
 3. The minimum distance between 2 revolving doors or between points A or B and a revolving door is 3 m.
- (c) Form an inequality for the number of revolving doors, n , that can be built along AB in the form $a < n < b$, where a and b are numbers.
- (d) Hence, write down the number of revolving doors that could be built, giving a reason for your answer.

Application of Mensuration in Real World context (Secondary 2 Topic)

The gherkin is an iconic commercial building in London City. The building can be modelled as a cylinder with a cone as its top.



The height of the conical top is 10 m and the radius of the cylindrical building is 20 m. There is a total of 61 floors in the building. Each floor is circular in design. The outer layer of the circular floors of the building can be depicted below.



The layout of the glass triangular windows is such that the next window is an inverted form of the previous one. Any unused spaces between the triangular window panes are sealed with a sealant of negligible weight.

- a) Determine the maximum even number of windows that can be installed at each floor.

Every floor is separated with 2 cement layers with height of 0.5 m each and triangular windows. The mass of the conical top is 2400 kg. The total length of steel bars used for the Gherkin in construction is 35 km. Each floor can hold a minimum mass of office equipment up to 300 kg per m² of floor area.

The following materials used and their respectively masses for the Gherkin's construction are given in the table below:

Material	Mass (kg)
Glass window	20 kg each
Cement	10 kg/cubic metres
Steel	2.5 kg/m

- b) The ground foundation laid for the Gherkin can support a total mass of 2.5×10^7 kg. Determine if the ground foundation is enough to support the building with maximum load. Justify your answer with relevant working.

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Applications of Arithmetic in Real-World Context (Secondary 3 Topic)

To help alleviate financial problems caused by the COVID-19 pandemic, Deputy Prime Minister Heng Swee Keat introduced the S\$48.4 billion Resilience Budget. This follows the S\$4 billion Stabilisation and Support Package for workers and firms, which was part of the S\$6.4 billion Unity Budget.

- (a) (i) Express 48.4 billion in standard form.
(ii) What is the total amount of money the Singapore government has placed in so far to help alleviate the financial problems Singaporeans may face. Give your answer in millions.



RESILIENCE BUDGET FOR COVID-19 PANDEMIC

HELP FOR THE SELF-EMPLOYED

The Singapore Government will provide direct cash assistance to more self-employed people, to help them tide over this difficult period

- Introduce a **S\$1.2 billion SEP Income Relief Scheme (SIRS)** to provide eligible self-employed people with **S\$1,000** a month for 9 months
- Extend the SEP Training Support scheme to Dec 2020
- Enhance the SEP Training Support scheme's hourly training allowance from S\$7.50 to **S\$10**, with effect from May 1, 2020



- (i) Using the information above, estimate the number of eligible self-employed people the Resilience Budget will be able to help? Give your answer to the nearest whole number.
(ii) As of September 2019, there is a total of 4026200 Singaporeans and Permanent residents. What is the percentage of Singapore Residents who will receive aid from the Resilience budget?

As part of the utilities rebate given to Singaporeans, all households with at least 1 Singaporean citizen will receive a S\$100 Solidarity Utilities Credit. In addition to that, they will also receive further rebate depending on their housing type which is illustrated in the table on the next page.

Housing Type	Rebate received in April 2020	Rebate received in July 2020	Rebate to be received in October 2020
1 – and 2 –Room	\$300	\$300	\$100 or \$200
3 –Room	\$270	\$270	\$90 or \$180
4 –Room	\$240	\$240	\$80 or \$160
5 –Room	\$210	\$210	\$70 or \$140
Executive/Multi Generation	\$180	\$180	\$60 or \$120

	Adam's family	Sherman's family
Number of working adults (inclusive of both parents who are working)	2	3
Number of seniors	2	0
Number of children below 21	2	2
Type of household	5 –room HDB flat	Executive Mansionette

The table above shows the number of people living in Adam and Sherman's household.

Find out which set of family will receive more aid and by how much more by September 2020.

Both sets of family are not considered as lower income and are not considered as large families as well.

Application of Statistics in Real World context (Secondary 4 Topic)

Mr Lim bought a new car with an engine capacity of 1599 *cc* for \$ 119 888. The specifications of his car as stated by the manufacturer are shown below.



Engine Capacity: 1599 *cc*
Acceleration: 11.1 *s* (0 – 100 *km/h*)
Top Speed: 185 *km/h*
Fuel Capacity: 55 *litres (L)*

He started driving with a full tank of fuel. After travelling 200 *km*, he discovered that he needed 30 *litres* of fuel to fill up the fuel tank completely.

- (a) Find the amount of fuel used in litres for 100 *km* of his journey.

He is keen to find out whether the fuel efficiency was indeed 15.3L/100 *km* as he claimed. He recorded 10 readings as shown below.

15.6 14.9 15.0 14.9 15.4 15.7 15.4 15.6 14.5 15.9

- (b) Calculate the mean and standard deviation of these 10 readings.
(c) Hence justify whether the manufacturer's claim is reasonable.

He paid a 50% down payment and took a car loan at 2.08% compounded annually for 5 years.

- (d) Calculate the amount of his car loan.
(e) Calculate the monthly instalment for his car.

The table below shows part of the road tax formula that is payable every six months in relation to the engine capacity of the vehicle. The road tax differs for cars running on **petrol** or **diesel**.

Engine Capacity	Vehicles running on petrol	Vehicles running on diesel
$1000\text{ cc} < EC \leq 1600\text{ cc}$	$[\$250 + \$0.375 \times (EC - 1000)] \times 0.6256$	$[\$250 + \$0.375 \times (EC - 1000)] \times 0.782$
$1600\text{ cc} < EC \leq 3000\text{ cc}$	$[\$475 + \$0.75 \times (EC - 1600)] \times 0.6256$	$[\$475 + \$0.75 \times (EC - 1600)] \times 0.782$
$EC > 3000\text{ cc}$	$[\$1525 + \$1 \times (EC - 3000)] \times 0.6256$	$[\$1525 + \$1 \times (EC - 3000)] \times 0.782$

- (f) Calculate the road tax payable by Mr Lim for his Toyota Altis which runs on petrol.

Mr Lim has another van which runs on diesel.

- (g) He recently paid \$ 2363.986 to renew the yearly road tax for the van. Calculate the engine capacity of the van.

(h) Suggest a reason why vehicles running on diesel have to pay a higher road tax.

The fuel prices and discounts given for the different types of fuel are shown in the table below.

Types of Fuel	Price per litre of fuel	Discounts if any
F1	\$2.43	14% off (Additional 5% cash rebate on nett purchase)
F2	\$2.09	15% off
F3	\$2.68	20% off
FC	\$1.50	2% discount (Excluding 3% carbon tax imposed)

(i) Suggest a sensible decision that Mr Cheong would make in deciding the type of fuel to use for his car. Justify your answer and show your calculations clearly.

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